



Rats drive tiny cars

Scientists have found that learning to drive small cars helps rats feel less stressed.
<https://digit.in/nov19-35>



Fungi over fertilizer

Fungi boosted the uptake of key nutrients in wheat and could lead to new, 'climate smart' crops. <https://digit.in/nov19-36>

BIOLUMINESCENCE IN NATURE

A detailed look into
nature's own glow sticks

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ave you ever come across the natural wonder of bioluminescence speckled across our planet? It's the

process by which animals, fish and insects produce illumination, creating some beautiful sights.

Bioluminescence can be described as the production and emission of light by living creatures in their bodies. The phenomenon, while gorgeous, isn't as rare in marine animals as you'd think. A whopping 76 per cent of ocean animals such as fish, bacteria, jellies and more, exhibit bioluminescence. These marine species produce their own light as a by-product of a series of chemical reactions in their body, or even by producing host bacteria that provide the illumination.

CHEMISTRY

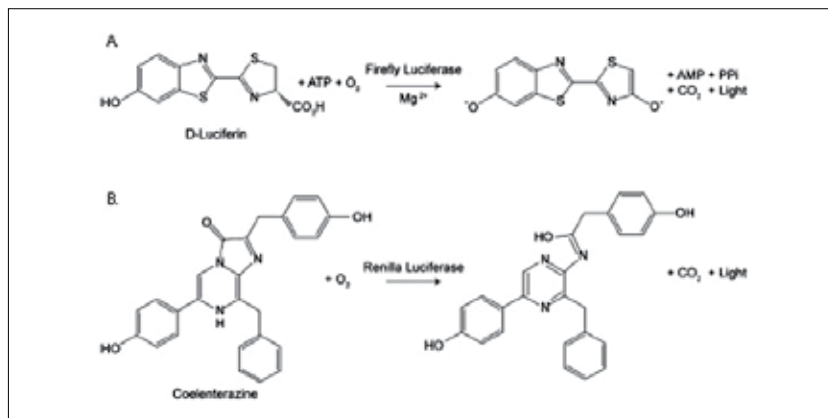
Researchers have also found that bioluminescence is a form of chemiluminescence. It produces blue-green light, also known as oxyluciferin, because of

the oxidation of a compound known as luciferin or the 'light-bringer' by an enzyme called luciferase. Bioluminescent creatures can usually produce or synthesise luciferin by themselves. For example, dinoflagellates, a type of plankton, produce a bluish-green colour through the process of bioluminescence occurring within their bodies.

However, some bioluminescent creatures do not synthesise luciferin but instead, they absorb it via other organisms, as either food or when

they are in a symbiotic relationship. For example, some species of midshipman fish acquire luciferin from the 'seed shrimp' they ingest. Some squids house bioluminescent bacteria within their light organs and their relationship with the bacteria is an example of a symbiotic relationship ending up in bioluminescence.

Bioluminescent dinoflagellates create light by the luciferin-luciferase reaction. Interestingly, the luciferase found in these dinoflagellates



Luciferin-Luciferase reaction in a firefly